

Distractions intervention strategies for in-vehicle secondary tasks: An on-road test assessment of driving task demand based on real-time traffic environment

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ABSTRACT

When driving a vehicle, the driver must allocate adequate attention to the demands of driving in order to be safe. Based on analyses of vehicle data, driving environment data, and videos of the road ahead, a driving task demand prediction model based on real-time road traffic data was established in this study. Assessments of driving task demands allowed for the validation of this model. In addition, intervention strategies were proposed for distractions from in-vehicle secondary tasks at different levels of demand. The analysis showed that at a high driving task demand, the control of all in-vehicle information systems (IVIS), except for the playing of radio or CD, should be warned against or forbidden. Meanwhile, distractions from secondary tasks originated from 91% of the in-vehicle foreign objects, and 67% of the in-vehicle facilities can be avoided by formulating precaution strategies. This study provides methods and technical support for the management of driver distraction precaution.

1. Introduction

1.1. Background

The driver of a motor vehicle generally performs a variety of tasks, both those directly involving driving and those that don't. Usually, the primary task is to steer the car and watch out for potential road hazards. Secondary tasks refer to those in-vehicle tasks that ask the driver to divide individual attention to perform. Such tasks include visual-related tasks, such as checking a map, responding to cell phone texts, or looking for items, as well as non-visual-related assignments, such as calling on the phone, listening to a radio programme, drinking a beverage, or thinking. Both types of tasks directly compromise the driver's visual ability, thus causing a driving distraction (Birrell and Young, 2011). A study from the National Highway Traffic Safety Administration (NHTSA, the United States) showed that out of the traffic collisions caused by driver distraction, about 30% (1 million cases) are caused by distractions caused by secondary tasks (Neville and Haddington, 2010). IVIS are also an important source of distraction (Greenberg et al., 2003). In-Vehicle-tasks will affect the driver's response time to the danger, and the braking time and acceleration will be higher (Nowosielski and Trick, 2018). Reducing driving mistakes and traffic accidents, as well as searching for effective approaches to eliminate the potential hazards in those tasks, are of significant, practical importance for the improvement of road traffic safety and the application of IVIS technology in the car industry (Regan, 2005).

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1.2. The current study

The latest surveys show that among various in-vehicle secondary tasks, using the audio entertainment system is performed by 70% of drivers. Using a cell phone or other wireless devices is the primary cause of accidents (Brodsky and Slor, 2013; Consiglio et al., 2003), followed by talking with passengers, comforting children in backseats, and other passenger-related behaviors are the second most frequent cause of accidents (Arnett et al., 1997; Regan et al., 2005). Using in-vehicle systems to receive and send e-mail greatly weakened driver's distraction and decreased their ability to control the vehicle (Xian and Jin, 2014). In-vehicle Tasks that require many glances and a high proportion of long glances away from the road are of special concern for safety (Tivesten and Dozza, 2014). The driver must allocate attention to the demand of driving tasks in order to drive safely (Engstrom and Johansson, 2005). With real-time assessment of the driving task demand, discouraging or forbidding some secondary tasks when the demand exceeds a pre-defined limit could help reduce distractions.

Driving task demand assessment was first introduced in studies of general in-vehicle driver subsidiary (GM Intelligent Driver Systems, GIDS) and in-vehicle multi-media communication (COMMUNICAR) projects. The GIDS project applies the multi-resource attention allocation theory to assess the workload from driving tasks, such as steering, overtaking, and secondary tasks. Based on information estimation and the assessed workload, the priority of each secondary task is determined (Rantanen and Goldberg, 1999). Multi-resource theory was proposed by Wickens and Knight et al., which assumes that human processing resources are three dimensions of a cubic structure with perceptual channels, coding, and stages. The perceptual channel includes visual and auditory levels. The stage includes three levels of sense, cognition, and response. The coding dimension is divided into two levels: spatial coding and speech coding. In addition, the attention of the visual channel to the information can also be divided into the focal point and the periphery so as to constitute the fourth dimension. If two tasks have a common demand level in one or more dimensions, time sharing will become poor and driving performance levels will decline. The car multimedia communication (COMMUNICAR), using driving data including brake and throttle positions, and Neural Networks Model evaluates the driving workload and judges the driver distraction. The Safety Vehicles Using Adaptive Interface Technology (SAVE-IT) research, subsidized by the NHTSA, seeks to manage vehicle-mounted systems through real-time monitoring of driver behaviors and road conditions (Witt et al., 2004; Zhang et al., 2006). However, the Adaptive Integrated Driver-vehicle Interface (AIDE) developed within the European Commission evaluated the type of driver distractions using different variables such as surface friction, driver predilection, and driver conditions; AIDE can provide warning information for use in newer in-vehicle technology and portable equipment (Peng et al., 2014).

At present, two common methods have been used to reduce the driver distraction issue (Horberry et al., 2006). One of methods adapts to a nonadaptive way, and the second method is adaptive interface technology. Through the first way, the distraction potential of non-driving tasks (such as editing text message) can be evaluated by models, performance index, and simulation procedures, which were based on test trajectory assessment, task analysis, laboratory experiments, and field research (Tango and Botta, 2013; Liang and Lee, 2014). Those three factors can help an experimental designer to better design a less distracting task (for example, designing commonly used text messages). Besides, a threshold line can be drafted to distinguish the tasks that are high-distracting or low-distracting. When the vehicle is moving, some less distracting tasks (such as radio tuning) may be allowed to operate, but high-distracting tasks may be prohibited. Some new principles have been proposed to assist drivers avoid distraction while improving the utility of on-board systems (Jin et al., 2012).

The attention level required by the driver is usually constrained by the driver's own conditions and the influence of different driving environments. Therefore, the non-adaptive method has certain limitations. The second method is adaptive interface technology, also known as workload management method, which can change dynamically with the different driving environment. This method is used to evaluate the attention needs from the driving task and the level of distraction from non-driving tasks through the pre-installing sensors in the vehicle. It tries to ease the driver's distraction by intervening or preventing. Through intervention strategy, when the driver showed more distraction than the driver's threshold, the distraction alert would inform the driver to redirect his attention to the driving task. Through prevention strategy, if the non-driving task has a trend to produce over certain threshold level of distracted driving conditions, the system could prevent the non-driving task, or issue advices to discourage the drivers involved in this non-driving task. In addition, when the adaptive method is adopted, the threshold of distraction depends on the driving condition, and the user interface can be modified.

Currently, there are few reports of studies in China related to in-vehicle secondary tasks. Most researchers focus on distraction sources outside the vehicles when studying driver attention allocation. Although no thorough or systematic studies of the relation between driving tasks and distractions from in-vehicle secondary tasks have been done, existing studies have established an important basis for this work. Based on previous researches, we concentrate on the intervention strategies to mitigate driver distraction, develop and evaluate driving task demand. The demand of driving task is tested by the variables collected by the in-vehicle sensors. Two strategies are discussed to reduce the driving distraction: The intervention strategy can provide the driver with demand-based feedback, while the prevention strategy avoiding certain non-driving tasks and providing need-based advice. Our strategies are aimed at preventing drivers from undertaking in certain non-driving tasks. Considering that many traffic accidents are caused by distractions from in-vehicle secondary tasks, there is an urgent need to study such distractions.

1.3. Aims and objects

On the basis of analyzing the demands of driving tasks and in-vehicle secondary tasks, this paper proposes intervention strategies for secondary task distractions based on the demand of driving tasks. Combining experimental data, the driving task demand is assessed and predicted. A driving task demand prediction model based on real-time traffic data is established. The outcome of the

research will support to the early warning and intervention of driver distractions.

2. Method

2.1. Participants

At present, young people and middle-aged people use more driver assistance systems, while older people accept less. Therefore, the object of this paper is from 20 to 50 years old. 20 healthy volunteers (the same ratio of male and female) aged 22–46 years old ($S.D. = 4.27$, $M = 27.30$) participated in the experiment. Distracted driving is a ubiquitous phenomenon. In the previous studies proved that there was no significant difference between the young drivers and the older drivers (Parnell et al., 2018; Pope et al., 2017). So our study did not have a specific requirement for age distribution. All the volunteers held valid driver's licenses, were insured and were skilled drivers (at least 5 years driving experience; $M = 8.32$, $S.D. = 4.19$) who drove 5000–10,000 km annually. All the tested drivers were in good physical and mental health, without visual and auditory disorder, as well as the habit of drinking and taking drugs. Drivers need to be investigated their basic information and then filled in the questionnaires before the experiment. The experiments were conducted during both rush and non-rush hours, also considered during the daylight factors and nighttime factors. All participants drove between 9 a.m. and 3p.m. (daylight conditions) with 6.30p.m. to 9.30p.m. (nighttime conditions). The survey included peak hours and non-peak hours. Participants were informed that they could stop participating in the experiment at any time without any penalty, all participants gave their informed consent.

2.2. Experimental equipment

In this study, simulations and field tests were performed to analyze the demand of driving tasks. Experimental equipments were installed on the test vehicles, including several cameras (front camera, lane tracking camera and others), a remote front radar detector, a steering angle sensor, a yaw-rate sensor, a brake sensor, and a turning signal activator. The radar detector can detect at most 20 targets within 150 m ahead, and determine the relative distance, relative speed, and angle of the target. Based on the relative speed, it can be determined whether the target is moving at the same speed as the test vehicle. The front camera was mainly used to monitor sudden pedestrians in front of the vehicle, which can eliminate the blind spot of sight range, and also serve as a warning function. Steering angle sensor was used to measure the rotation angle of a steering wheel when turning to the vehicle. Mostly it was a non-contact multi-turn absolute angle measurement. Yaw-rate sensor was responsible for the determination of the vehicle around the vertical axis rotation movement. When the car dodge obstacles in an emergency, or cornering occurs in understeer, over-steering, it can help the vehicle to overcome the tendency to deviate from the ideal trajectory. The brake sensor recorded the driver's pressure on the brake pedal. When the sensor detected a large pressure shock, the system would start emergency braking. A Safe TRAC lane tracker was used to measure the driver's lateral lane position and changes, and determine the vehicle's lateral position and movement. With a laser distance measurer, the vehicle's real-time distances from surrounding targets, including other vehicles and pedestrians, were evaluated. With the AS1300 truck driving simulation system, different road traffic environments were constructed to provide visual, auditory, and cognitive feedback. The driver's biological indicators, eye movements, judgment errors, and operational errors were recorded to evaluate the driver's distraction and driving performance. The distribution of the cameras on the test vehicle is shown in Fig. 1.

2.3. Design and procedure

After being explained the procedure, collecting the driver information and filling in the consent form Participants familiarized the vehicle environment by driving a training route for 15 min. The test route was included both city roads and highways, and the experiments were conducted during both peak and non-peak hours. During the experiments, vehicle speeds, relative distances, relative speeds, and other data were collected, in addition to the road traffic environment ahead of the vehicle. From the collected

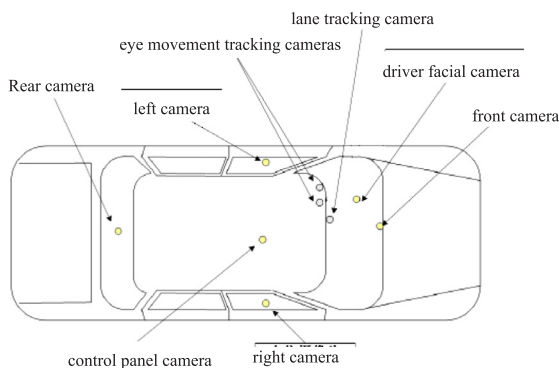


Fig. 1. Video camera distribution in the test vehicle.

video data, 100 video recordings with different surrounding environments varying in degrees of complexity were selected. The selected traffic environments included zero or very low traffic volume, as well as high traffic volume. The selected videos also included different vehicle movements, such as lane changes, merging, or splitting. Each video is about 8 s in length, a duration chosen to avoid recording changes in the degree of complexity of the traffic environment. Meanwhile, the sensors were also given enough time to collect reliable data for subjective assessment of the driving task demand.

3. Driving task demand assessment and prediction

3.1. Assessment of driving task demand

It's necessary for the experimenters to sit in the driving simulation seat and watch the 100 video recordings to assess the driving task demand. Each subject was asked to give a score of 1–5 to the traffic environment reflected in each video (1 means low demand, 3 means medium demand, 5 means high demand). According to the attention required to monitor cases and risks and to control the lane position and driving speed, so as to avoid collisions or rush out of road traffic accidents.

Considering the differences in the assessments of the 20 subjects, an 85% confidence interval was chosen. The mean of the samples was accustomed to represent the driving task demand in a certain road traffic environment. The scores were divided into three types:

- (1) Low demand driving tasks, with average score equals to or smaller than 2.25 (15 video recordings);
- (2) Medium demand driving tasks, with average score greater than 2.25 and smaller than 3.50 (65 video recordings);
- (3) High demand driving tasks, with average score equals to or greater than 3.50 (20 video recordings).

3.2. Prediction of driving task demand based vehicle sensor data

Each prediction of the driving task demand was made based on eight seconds of video. The demand is associated with several variables, including the average headway and the test vehicle speed. If the speed of test vehicle relative to the ahead one is zero, it means the target is moving towards the test vehicle with the same speed. If the relative speed is positive, the target is moving faster than the test vehicle, and a negative speed indicates the target is moving slower or is moving to the contrary path. If the target is on the right to the test vehicle, the angle is a positive value, and otherwise, it is a negative value. Other variables less related to the driving task demand are target speed and the speed of the vehicle ahead. Based on those variables associated with the driving task demand, a correlative factor prediction model as follows has been established.

$$W = \sum_{i=1}^n \left(\frac{k}{\left(1 + \frac{|\theta_i - \beta|}{4}\right) \min\left(\frac{d_i}{25}, \frac{d_i}{v}\right)} \right) \quad (1)$$

where W is the driving task demand, k is a constant, θ_i is the horizontal angle between the test vehicle and the target, β is the angle of the target, d_i is the distance of the test vehicle to the target (m), and v stands for the speed of the test vehicle (m/s). If the target is moving at the same direction as the test vehicle, $k = 1$; if the target is moving in a different direction, $k = 4$; and if the target is not moving, $k = 0$.

As Eq. (1) shows, d_i and the minimum value of d_i/v (i.e. headway) reflect the driving task demand. As d_i and the headway decrease, the driving task demand increases. Along with the increase of the number of targets, the driving task demand also increases. The model is influenced by the factors in the following equation:

$$W' = f_r \cdot f_b \cdot f_v \cdot f_p \cdot W \quad (2)$$

where W' is the modified value of the driving task demand. f_r is the yaw rate; when the yaw rate exceeds a specific value (e.g., $f_r > 4^\circ$), it indicates that the vehicle is driving on a curved road, and the model needs to be modified; a large yaw rate (e.g., $f_r > 10^\circ$) indicates acute turns or U-turns, and the model needs intensive modification. f_b is the lane width modification coefficient; when the lane width is smaller than the threshold (3.25 m), the value of W needs to be up-regulated. f_v is the vehicle speed modification coefficient; when the test vehicle's speed is lower than the threshold (15 m/s), the value of W needs to be down-regulated. f_p is the brake coefficient; when the brake is working, the value of W needs to be re-estimated.

According to the cumulative frequencies, the demand less than 15% of the total data is considered to be the low demand, not less than 15% but less than 80% of the total data is considered to be the medium demand, and more than 80% considered to be the high demand.

The modified driving task demand W' is divided into three types as below:

- (1) I:Low demand ($W' < 0.5$)
- (2) II:Medium demand ($2.26 > W' \geq 0.5$)
- (3) III:High demand ($W' \geq 2.26$)

Fig. 2 shows the relationship between the assessed and predicted driving task demands. A correlation coefficient of 0.83 was

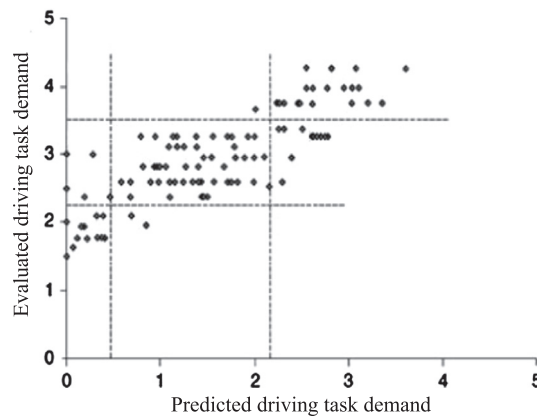


Fig. 2. Estimated and predicted levels of driving task demands.

obtained, indicating that the proposed real-time dynamic demand prediction model is effective.

As seen in Fig. 2, 10 samples with high predicted driving task demand levels were assessed as medium. Among those 10 samples, eight samples had driving task demand levels of 3.25–3.40, which were very close to high demand levels (average grade ≥ 3.3). For the video recordings with low predicted demand levels, most records had assessed level of 2.4, which was very close to the low demand level. Table 1 shows a cross reference of the predicted demands and assessed demands.

As seen in Table 1, for 83% of the video recordings, the assessed and predicted demand levels are consistent. For the other 17%, the assessed and predicted demand levels are not the same (for example: the level of demand prediction is medium, while the level of demand assessment is low). There is no case with large variation between prediction and assessment (e.g., predicted demand level is high while assessed demand level is low).

The assessed demand levels in Fig. 2 and Table 1 are average values of the subjective assessments of 20 drivers. Thus, the consistency between predicted values and subjectively assessed values reflect the driving task demand levels in specific road traffic environments.

3.3. Attention allocation to in-vehicle secondary tasks based on driving task demand

The visual scanning range of the driver is associated with the position of eye focus and the head. When the driver's eye focus or head position is in the rectangular angle range of $24^\circ \times 24^\circ$, it can be thought that the driver's scanning range located on the road ahead. Otherwise, it is considered that the driver is looking at other targets besides the road. Our study showed that if the driver's visual scan range moves from the road exceeds a certain threshold, indicators of distraction like lane position deviation, lane departure, and brake time increase extensively. Driving performance also declines with time that the eye is not focused on the road. Therefore, when the driver's visual scan range has been off the road for a specific time period, the driver should be given notice.

In this study, the driving task demand prediction is based on real-time road traffic conditions. When the predicted demand is greater than the threshold, corresponding precaution needs to be taken. Certain in-vehicle secondary tasks should be recommended or forbidden in order to reduce driver distraction. For each video recording, the subjects were given questionnaires which asked to determine whether reducing driver distractions is needed in the specific driving environment. A secondary task that can be executed was given the value 1, a secondary task that is not recommended was given the value 2, and a secondary task that should be forbidden was given the value 3. The secondary tasks included adjusting the radio (looking for or turning buttons), playing the radio or CDs, replacing a CD, choosing MP3 items, reading navigator directions, and using an audio route guidance system. Fig. 3 shows the scores for each secondary task at different types of driving task demands.

As Fig. 3 shows, adjusting the radio and playing a CD are at similar recommendation levels. At medium driving task demand, these two tasks are scored 1 (allowed) or near 1 on average. When located at high driving task demand, the average scores are near 2 (not recommended). When located at low driving task demand, the recommendation level of MP3 is near 1 (allowed). The score is near 2 and over 3 when the driving task demands are medium and high, respectively. A clear relation is shown between driving task

Table 1
Cross reference between assessed and predicted driving task demands.

Demand prediction	Demand assessment		
	III	II	I
III	19	10	0
II	1	51	2
I	0	4	13

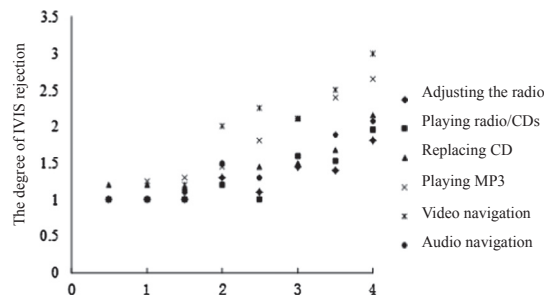


Fig. 3. Recommendation levels of secondary tasks at different levels of driving task demands.

demand levels and using a cellphone or a navigation system. The initial average value of inputting navigation information is 1.5, and then it reaches 2 (not recommended). When located at medium driving task demand, the average score for inputting navigation information is over 2, and when driving task demand is high, the average score for this task is almost 3 (forbidden). Reading directions from navigation systems got a score of 2 (not recommended) at medium level of driving task demand, and a score of 3 (forbidden) at high-level driving task demand. For audio navigation system, when lied in high or medium driving task demand, the score is about 2. This indicates that watching video navigation is more distracting than listening to directions. Therefore, navigation systems with a voice function are recommended in order to reduce driver distractions.

4. Results

4.1. Intervention strategies for distractions from in-vehicle secondary tasks

The demands of driving task is predicted by real-time road traffic condition. If it exceeds the threshold, it will not encourage or prohibit some secondary tasks to reduce driver distraction.

Table 2 illustrates the intervention strategies for distractions from IVIS at different levels of driving task demands. As can be seen, the warning against or forbidding of certain secondary tasks is determined by the driving task demand level.

When reach to high driving task demand, the use of all devices, except the radio and CD players, should be warned against or forbidden, especially the operation of visualized navigation systems and portable audio plays, which should be forbidden.

4.2. Intervention strategy for distractions from other in-vehicle secondary tasks

According to natural driving and experimental test data, combined with literature research conclusions about in-vehicle secondary tasks (Michael et al., 2009), the intervention strategy was given for distractions from other in-vehicle secondary tasks. Using the expert evaluation method, five potential categories in-vehicle secondary task distractions (Foreign Objects, Passengers, In-vehicle activities, In-vehicle moving objects and In-vehicle facilities) were evaluated to determine whether the impacts of some secondary tasks can be prevented. 20 experts were selected for evaluation, and when each Target/Event received more than 60% of the votes, the Target/Event was deemed to be avoidable. Table 3 reflected the results of evaluation.

The evaluation results showed that among the sources of in-vehicle distractions, 91% of the foreign objects in car, 67% of the in-vehicle facilities, 50% of the distractions from passengers, 50% of the in-vehicle object movements, and 57% of in-vehicle activities are avoidable. Therefore, for some in-vehicle sources of secondary tasks, strategies and approaches can be formulated to avoid.

Table 2

Intervention strategy for distractions from IVIS tasks at different levels of driving task demand.

In-vehicle secondary tasks	Driving task demand		
	I	II	III
Playing radio/CDs	No suggestion	No suggestion	No suggestion
Adjusting radio/CDs	No suggestion	No suggestion	Warning
Replacing CD	No suggestion	No suggestion	Warning
Audio navigation	No suggestion	No suggestion	Warning
Video navigation	No suggestion	Warning	Lockout
Playing MP3	No suggestion	Warning	Lockout

Note: Warning, An attention of the use of a certain device functions at a specific driving task demand.

Lockout, The ability to lock a certain device under specific driving task demand.

Table 3
Evaluation of potential in-vehicle secondary tasks.

Sources	Target/Event	Avoidable?	Sources	Target/Event	Avoidable?
Foreign objects	Animals	✓	In-vehicle activities	Daydreaming/Thinking	—
	Files	✓		Drug effects	✓
	Water	✓		Singing	✓
	Dinnerware	✓		Talking	✓
	Food	✓		Sneezing	—
	Decorations	✓		Coughing	—
	Lost items	—	In-vehicle moving objects	Animals	✓
	Self-impact	✓		Insects	—
	Cigarettes	✓		Stationary objects	—
	Writing tools	✓		Personal objects	✓
	Technology devices	✓		Entertainment devices	✓
			In-vehicle facilities	Mechanical problems	—
Passengers	Adults	✓		Vehicle control facilities	✓
	Children	—			

* In Table 3, the symbol “✓” indicates avoidable, and the symbol “—” indicates unavoidable.

5. Conclusions

Driving a vehicle is a multi-level task consisting of various driving tasks and secondary tasks. The driver must allocate attention equivalent to task demands in order to drive safely. Combining experiments and tests, this paper presents the following conclusions.

- (1) Based on vehicle data and real-time driving environment data as well as videos of roads ahead, a driving task demand prediction model has been established. Through assessment of the driving task demand levels, the established model has been validated.
- (2) Combining analysis of driving task demands in actual road traffic conditions, an intervention strategy was proposed for the distractions from secondary tasks at different levels of driving task demands. For example, at high levels of driving task demand, the driver should allocate more attention to driving tasks. On the other hand, at low levels of driving task demand, the driver could allocate part of attention to specific secondary tasks.
- (3) When the driving task demand exceeds a pre-defined value, the measures should be taken. Some secondary tasks are discouraged or forbidden, in order to reduce driver distraction in such condition.
- (4) Some secondary tasks derived from 91% of in-vehicle foreign objects and 67% of in-vehicle facilities are avoidable. Thus, corresponding countermeasures such as setting specific rules can be taken to avoid such distractions.

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